

Generalized Reed Solomon Codes (GRS Codes).

Let $\mathbb{F}_q$ be a finite field.

→ $\alpha_1, \alpha_2, \dots, \alpha_n$ be distinct nonzero elements of $\mathbb{F}_q$. ($n \leq q-1$).

→ $v_1, v_2, \dots, v_n$ be nonzero elements (not necessarily distinct)

→ Consider $(n-k) \times n$ matrix as follows:-

$$H_{GRS} = \begin{bmatrix} v_1 & v_2 & \dots & v_n \\ v_1 \alpha_1 & v_2 \alpha_2 & \dots & v_n \alpha_n \\ v_1 \alpha_1^2 & v_2 \alpha_2^2 & \dots & v_n \alpha_n^2 \\ \vdots & \vdots & \ddots & \vdots \\ v_1 (\alpha_1)^{n-k-1} & v_2 (\alpha_2)^{n-k-1} & \dots & v_n (\alpha_n)^{n-k-1} \end{bmatrix}$$

GRS Code:- $\mathcal{C}_{GRS}$ is a code over $\mathbb{F}_q$

$$\mathcal{C}_{GRS} = \left\{ \underline{c} \in \mathbb{F}_q^n \mid H_{GRS} \underline{c} = 0 \right\}.$$

$\{\alpha_1, \alpha_2, \dots, \alpha_n\} \rightarrow$ are called as
Code locators or Column
locators.

Elements $\{v_1, \dots, v_n\}$ are called column
multipliers.

Property ①

QRS Codes are MDS Codes

$(n, k, n-k+1)$ achieves
Singleton bound with equality.

Consider H_{QRS} and show that any $n-k$
Columns of H_{QRS} are linearly independent.
 $\Rightarrow d_{min} = n-k+1.$

[Vandermonde matrix and there is explicit
formula for the determinant of Vandermonde
matrix.]

GRS Codes are MDS Codes.

To prove this, we will show that any $n-k$ columns of H_{GRS} form a full rank matrix.

Vandermonde Determinant Formula

$$\det \begin{bmatrix} 1 & 1 & \dots & 1 \\ \beta_1 & \beta_2 & \dots & \beta_r \\ \beta_1^2 & \beta_2^2 & \dots & \beta_r^2 \\ \vdots & \vdots & \ddots & \vdots \\ \beta_1^{r-1} & \beta_2^{r-1} & \dots & \beta_r^{r-1} \end{bmatrix}_{r \times r} = \prod_{1 \leq i < j \leq r} (\beta_j - \beta_i)$$

If $\beta_1, \beta_2, \dots, \beta_r$ are distinct elements in the field $\mathbb{F}_q$, then determinant is non zero.

Proof is by induction.

$r=2$ (base case)

Assume induction hypothesis for $r-1$

Consider the following determinant.

$$\begin{vmatrix} 1 & 1 & & 1 & 1 \\ \beta_1 & \beta_2 & \dots & \beta_{r-1} & z \\ \beta_1^2 & \beta_2^2 & & \beta_{r-1}^2 & z^2 \\ \vdots & \vdots & & \vdots & \vdots \\ \beta_1^{r-1} & \beta_2^{r-1} & & \beta_{r-1}^{r-1} & z^{r-1} \end{vmatrix} = D(z).$$

$D(z)$ is a polynomial in $F_q[z]$
and $\deg(D(z)) = r-1$

Coeff of z^{r-1} in the above determinant is

$$z^{r-1} \begin{vmatrix} 1 & 1 & 1 \\ \beta_1 & \beta_2 & \beta_{r-1} \\ \vdots & \vdots & \vdots \\ \beta_1^{r-2} & \beta_2^{r-2} & \beta_{r-1}^{r-2} \end{vmatrix}$$

Coeff of $z^{r-1} \neq 0$ (By induction hypothesis).

Roots of $D(z) \rightarrow \beta_1 \dots \beta_{r-1}$ as the roots

$$(z - \beta_i) \mid D(z) \quad \forall 1 \leq i < r.$$

$$\prod_{1 \leq i < r} (z - \beta_i) \mid D(z) \rightarrow \deg^{r-1} \text{ polynomial.}$$

$\hookrightarrow \deg^{r-1} \text{ polynomial}$

$$D(z) = (\text{Const}) \prod_{1 \leq i < r} (z - \beta_i).$$

Constant = Coeff of z^{r-1} in $D(z)$.

$$= \begin{vmatrix} 1 & 1 & & 1 \\ \beta_1 & \beta_2 & & \beta_{r-1} \\ \beta_1^2 & \beta_2^2 & & \beta_{r-1}^2 \\ \vdots & \vdots & & \vdots \\ \beta_1^{r-2} & \beta_2^{r-2} & & \beta_{r-1}^{r-2} \end{vmatrix} = \prod_{1 \leq i < j \leq r-1} (\beta_j - \beta_i)$$

By induction hypothesis.

$$D(z) = \left(\prod_{1 \leq i < j \leq r-1} (\beta_j - \beta_i) \right) \left(\prod_{1 \leq i < r} (z - \beta_i) \right).$$

We have defined $D(z)$ by replacing β_r with variable z .

Determinant of Vandermonde matrix of size $r \times r$;

$$D(\beta_r) = \left(\prod_{1 \leq i < j \leq r-1} (\beta_j - \beta_i) \right) \left(\prod_{1 \leq i < r} (\beta_r - \beta_i) \right)$$

$$H_{GRS} = \begin{bmatrix} v_1 & v_2 & \dots & v_n \\ v_1 \alpha_1 & v_2 \alpha_2 & \dots & v_n \alpha_n \\ v_1 \alpha_1^2 & v_2 \alpha_2^2 & \dots & v_n \alpha_n^2 \\ \vdots & \vdots & \ddots & \vdots \\ v_1 \alpha_1^{n-k-1} & v_2 \alpha_2^{n-k-1} & \dots & v_n \alpha_n^{n-k-1} \end{bmatrix}$$

Any $(n-k)$ columns of H_{GRS} are linearly independent.

$$\begin{vmatrix} v_1 & v_2 & \dots & v_{n-k} \\ v_1 \alpha_1 & v_2 \alpha_2 & \dots & v_{n-k} \alpha_{n-k} \\ \vdots & \vdots & \ddots & \vdots \\ v_1 \alpha_1^{n-k-1} & v_2 \alpha_2^{n-k-1} & \dots & v_{n-k} \alpha_{n-k}^{n-k-1} \end{vmatrix} = \left(\prod_{i=1}^{n-k} v_i \right) \left(\prod_{1 \leq i < j \leq n-k} (\alpha_j - \alpha_i) \right).$$

All α_j 's are distinct nonzero elements.

All v_i 's are nonzero elements

Thus, $\dim$ of $\mathbb{C}_{GRS} = n-k+1$.

Dual of a GRS Code is also a GRS Code.

Proof :- G_{GRS} is the parity check matrix for the dual code

$$G_{GRS} = \begin{bmatrix} v_1' & v_2' & \dots & v_n' \\ v_1' \alpha_1 & v_2' \alpha_2 & \dots & v_n' \alpha_n \\ \vdots & \vdots & \ddots & \vdots \\ v_1' \alpha_1^{k-1} & v_2' \alpha_2^{k-1} & \dots & v_n' \alpha_n^{k-1} \end{bmatrix}$$

$$G_{GRS} \cdot H_{GRS}^T = 0$$

True for any generator matrix & parity check matrix.

If there exist $v_1', v_2' \dots v_n'$ all nonzero elements such that

$$G_{GRS} \cdot H_{GRS}^T = 0$$

then we have proved the statement.

$$\begin{bmatrix} v_1' \alpha_1^i & v_2' \alpha_2^i & \dots & v_n' \alpha_n^i \end{bmatrix} \cdot \begin{bmatrix} v_1 \alpha_1^l \\ v_2 \alpha_2^l \\ \vdots \\ v_n \alpha_n^l \end{bmatrix} = 0 \quad \forall i, l. \quad \begin{matrix} 0 \leq i \leq k-1 \\ 0 \leq l \leq n-k-1. \end{matrix}$$

$0 \leq i+l \leq n-2$

We will write $(n-1)$ equations in matrix form

⌞ (Why only $n-1$, because the equation is only dependent only on the value of $i+1$)

$$\begin{bmatrix} v_1 & v_2 & \dots & v_n \\ v_1 d_1 & v_2 d_2 & \dots & v_n d_n \\ v_1 d_1^2 & v_2 d_2^2 & \dots & v_n d_n^2 \\ \vdots & \vdots & \ddots & \vdots \\ v_1 d_1^{n-2} & v_2 d_2^{n-2} & \dots & v_n d_n^{n-2} \end{bmatrix} \begin{bmatrix} v'_1 \\ v'_2 \\ \vdots \\ v'_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ 0 \end{bmatrix}$$

$\uparrow$
 $H'_{GRS}(n, 1, n)$

$n-k-1 = n-2$

$H'_{GRS} v' = 0$

$[v'_1 \dots v'_n]$ belongs to the code.

↓ C'_{GRS} with parameters $(n, 1, n)$

→ a vector which belongs to the code is clear

Minimum distance of C'_{GRS} is n is telling you that the vector $[v'_1 \dots v'_n]$ has all nonzero elements.

Interpreting GRS Codewords as polynomial evaluations

$$G_{GRS} = \begin{bmatrix} v_1' & v_2' & \dots & v_n' \\ v_1' \alpha_1 & v_2' \alpha_2 & \dots & v_n' \alpha_n \\ \vdots & \vdots & \ddots & \vdots \\ v_1' \alpha_1^{k-1} & v_2' \alpha_2^{k-1} & \dots & v_n' \alpha_n^{k-1} \end{bmatrix}$$

$$\underline{u} = (u_0, u_1, \dots, u_{k-1})$$

$$\underline{c} = (c_1, \dots, c_n)$$

$$\underline{c} = \underline{u} G_{GRS}.$$

$$= \begin{pmatrix} u_0 & u_1 & \dots & u_{k-1} \end{pmatrix} \begin{bmatrix} v_1' & v_2' & \dots & v_n' \\ \vdots & \vdots & \ddots & \vdots \\ v_1' \alpha_1^{k-1} & v_2' \alpha_2^{k-1} & \dots & v_n' \alpha_n^{k-1} \end{bmatrix}$$

$$c_j = \sum_{i=0}^{k-1} u_i v_j' (\alpha_j)^i = v_j' \left(\sum_{i=0}^{k-1} u_i \alpha_j^i \right)$$

Message polynomial $\hookrightarrow$

$$u(x) = \sum_{i=0}^{k-1} u_i x^i$$

$$u(x) \longrightarrow (v_1' u(\alpha_1) \quad v_2' u(\alpha_2) \quad \dots \quad v_n' u(\alpha_n))$$

Special case of $v_j' = 1 \forall j$

$$u(x) \rightarrow (u(\alpha_1) \ u(\alpha_2) \dots u(\alpha_n)).$$

Code description of Reed Solomon Code
(Case when $v_j' = 1 \forall j$)

$$C_{RS} = \left\{ (u(\alpha_1), \dots, u(\alpha_n)) : \begin{array}{l} u(x) \in \mathbb{F}_{q^1}[x] \\ \text{of deg} \leq k-1 \end{array} \right\}.$$

Alternate Proof for the MDS property of RS codes (described in terms of polynomial evaluations).

Length of $C_{RS} = n$.

Dimension = k (?). Look at the message polynomial $u(x)$ and see how many free coeffs are there
 $u_0 + u_1 x + u_2 x^2 + \dots + u_{k-1} x^{k-1}$
 $u_i \in \mathbb{F}_{q^1}.$

$\mathbb{C}_{RS}$ is still a linear code

Minimum distance of a linear code
= minimum Hamming weight of a
nonzero codeword.

$$(u(\alpha_1) \ u(\alpha_2) \ \dots \ u(\alpha_n))$$

and see how many zeros are there
for some nonzero polynomial $u(x)$.

No. of zeros of ^{at most} deg $k-1$ polynomial
 $\leq (k-1)$

$\Rightarrow$ No. of zero coordinates in any nonzero
codeword $\leq k-1$

$\Rightarrow$ No. of nonzero coordinates in any
nonzero codeword $\geq n - k + 1$

$$d_{\min} \geq n - k + 1$$

$\Rightarrow d_{\min}(\mathbb{C}_{RS}) = n - k + 1$ (Singleton bound).